

Online-Adaptive Fuzzy Cognitive Maps for Neuro-symbolic Financial Forecasting

Abstract—Financial markets exhibit complex, nonlinear dynamics and regime shifts that challenge traditional econometric models. We propose a neuro-symbolic framework that integrates Fuzzy Cognitive Maps (FCMs) with a deep recurrent neural network (long short-term memory, LSTM) and real-time news sentiment analysis. The system maintains dual parallel FCM pipelines (Yahoo Finance and Polygon/Google) whose causal edges evolve online via an error-driven adaptation rule. Lexicon-based sentiment extracted from NewsAPI serves as an exogenous bias. The evolving FCM states feed into the LSTM, and a simple ensemble combines all forecasts. On NASDAQ Composite data, the learned-weight ensemble reaches 77.6% state-directional accuracy, on par with the standalone LSTM (77.7%) while the fuzzy cognitive maps alone reach 66%—well above chance—and add interpretable causal structure that the neural baseline lacks. The framework provides interpretable causal graphs that update dynamically, offering transparent and adaptive trading signals.

Index Terms—Fuzzy cognitive maps, online adaptation, financial forecasting, sentiment analysis, LSTM, ensemble learning

I. INTRODUCTION

Various heterogeneous factors and their interactions predict financial markets. These factors include market volatility, financial crises, firm-specific news, earnings, geopolitical events and so on [1]–[6]. They operate under incomplete information and behavioral biases. This contributes to the complexity of modeling such system. The time-dependency and change in these factors explains the dynamic nature of financial market modeling [7]–[9].

Traditional econometric frameworks such as autoregressive integrated moving average (ARIMA) [10]–[12], vector autoregression (VAR) [13], and linear factor models [14] assume stationary and linearity. But historical financial market data suggests otherwise. It shows that asset returns exhibit heavy tails, volatility clustering, nonlinear dependencies, and structural breaks that violate the linear and stationary assumption. Financial practitioners also need transparent models that are interpretable and reveal causal drivers because of the financial implications of decisions based on such predictions.

Most machine learning models for financial market prediction fail to combine the desirable properties in a single framework: handling heterogeneous data, causal structure, dynamic behavior, and interpretability. Some combine heterogeneous signals, technical indicators from news and media attention in neural based models such as LSTMs and transformers to improve prediction performance [15]–[17]. Most of them use post-hoc tools like SHAP and LIME to achieve explainability [18]. These neural-based methods excel at capturing long-range dependencies but they lack explicit causal structure and interpretability. Other prior works extract causal graphs

from financial news [19] and learn causal representations for robustness [20]. However, their causality is acyclic and static, interpretability is external to the predictor, and inputs or functional form are restricted.

Fuzzy Cognitive Maps (FCMs) provides an interpretable graph-based framework for modeling complex systems. FCMs are directed graphs that model the causal relationships in a system [21]. Their nodes denote concepts, and their signed, weighted edges quantify the direction and strength of each causal link. FCMs admit feedback loops and iterate to produce dynamic trajectories rather than static outputs for acyclic causal graphs [22]. FCM predictions trace directly to the signed edges that drove it and explainability is built into the mechanism rather than post-hoc recovery. This pairing of causal structure, feedback dynamics, and intrinsic transparency in one model has driven FCM adoption across complex decision domains [23], [24].

Some prior works have applied FCMs for financial analysis including expert-built maps of stock-market dynamics [25] to interval-valued maps learned with particle swarm optimization for stock-return prediction. The recent work CausalSeqGNN is a hybrid model that gives FCM nodes Fama-French-inspired fuzzy state vectors, learns the map with a genetic algorithm, and couples it with a graph neural network for cross-sectional forecasting [26]. These methods either elicit these maps from experts or learn them offline. After training, they hold the map fixed. This leaves two questions open. How can we build FCMs from heterogeneous multi-source feeds and literature-grounded LLM knowledge? And how can we adapt them online as the market moves?

We present two frameworks for neurosymbolic modeling with FCMs to address these questions. Framework I is a predictive hybrid model that uses FCM to extract explainable and informative features for LSTM prediction. It ingests five sources, Yahoo Finance and Google Finance prices, Yahoo News and Google News sentiment, and key macroeconomic indicators, builds an FCM from them, and ensembles the FCM with an LSTM for forecasting. Figure 1 shows the FCM pipeline for Framework I and the resulting causal map. Framework II is an explanatory model. It fuses two knowledge sources, peer-reviewed literature and LLM knowledge, under human oversight, and produces a transparent, verifiable map of how financial factors interact rather than a forecast. Figure 2 shows the FCM pipeline for Framework II.

Our work makes three main contributions:

- 1) We introduce fully online FCM adaptation that updates edges at each time step instead of in batch.

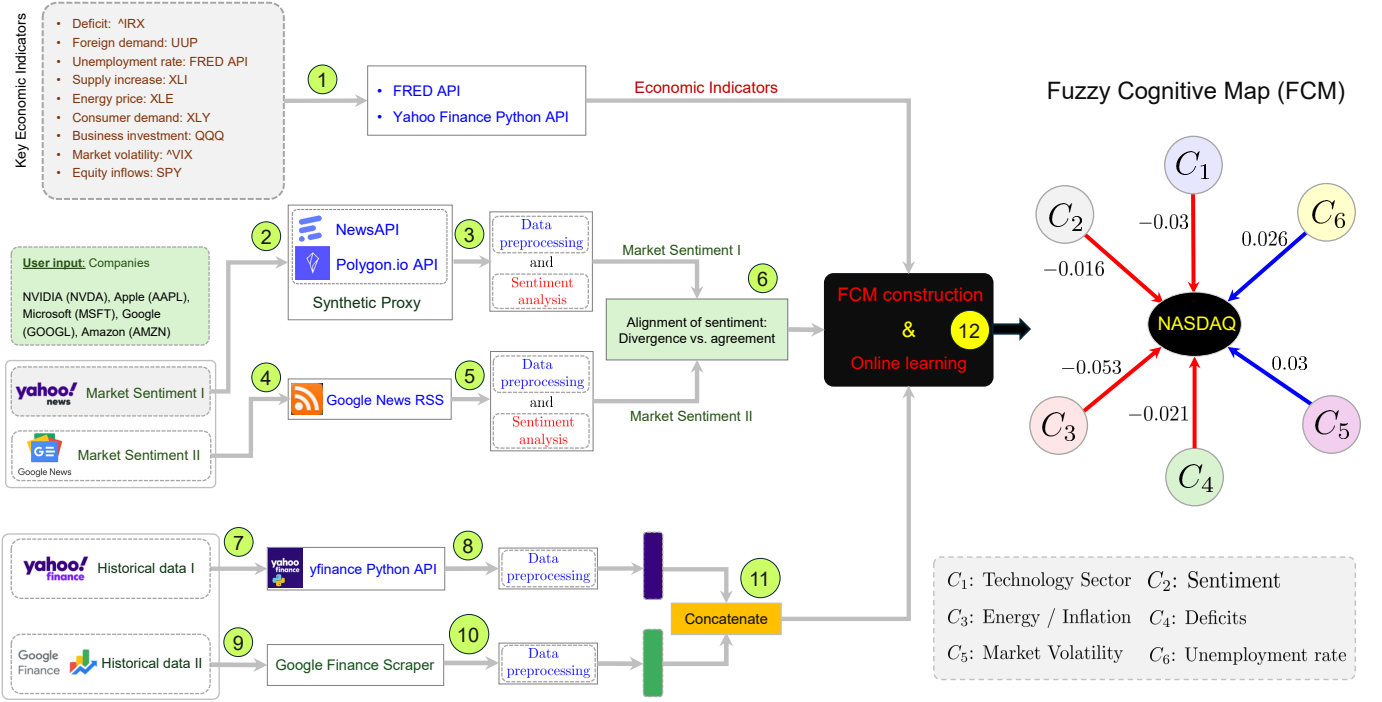


Fig. 1. Data pipeline and online FCM construction for Framework I: The system ingests three input streams and fuses them into an online-adapting fuzzy cognitive map. Step (1) pulls key economic indicators through the FRED API and the Yahoo Finance Python API and routes them directly to FCM construction. For news sentiment, the user specifies target companies (NVDA, AAPL, MSFT, GOOGL, AMZN); steps (2–3) query NewsAPI and the Polygon.io API as a synthetic proxy for Yahoo News, then preprocess the text and run sentiment analysis to produce Market Sentiment I. Steps (4–5) scrape Google News RSS and apply the same preprocessing and sentiment analysis to produce Market Sentiment II. Step (6) aligns the two sentiment streams and measures their divergence versus agreement. For price history, steps (7–8) retrieve Yahoo Finance data through the Yahoo Finance API and preprocess it. Steps (9–10) retrieve Google Finance data through a scraper and preprocess it, and step (11) concatenates the two price streams. Step (12) constructs the FCM and performs online learning, combining the economic indicators, aligned sentiment, and concatenated prices while updating edge weights as new data arrives. The right panel shows the resulting fuzzy cognitive map after online adaptation on NASDAQ Composite data (2020–2025). Blue arrows denote positive causal influence; red arrows denote negative influence. Edge labels show the final adapted weights $w_{i,NASDAQ}$. Energy price increase carries the strongest suppressive effect (-0.053), while low unemployment and elevated market volatility carry positive influence on NASDAQ.

- 2) We build a neurosymbolic ensemble that fuses FCMs from multiple data sources (Yahoo and Google prices, news sentiment, and macro indicators) with an LSTM.
- 3) We construct a literature-grounded, LLM-augmented FCM with per-edge citations and verifiable justifications, yielding an auditable map of how financial factors interact.

On NASDAQ Composite data, our learned-weight ensemble reaches 77.6% state-directional accuracy, on par with a standalone LSTM baseline (77.7%) while providing interpretable, dynamically updating causal graphs; the FCM components alone reach 66%, and a naive uniform average of the components reaches 70.4%. Residual analysis confirms that the remaining error is stochastic noise, indicating successful signal extraction. The adaptive FCMs converge to new market regimes within 20–30 trading days.

II. BACKGROUND AND NOTATION

A. Fuzzy Cognitive Map

An FCM is a directed graph $G = (N, E)$ with N nodes and edges $w_{ij} \in [-1, 1]$. The value of the w_{ij} indicates the strength and magnitude of the causal link from i^{th} node C_i to

the j^{th} node C_j . Positive weights indicate excitatory influence, negative weights means inhibitory influence, and zeros weights means no influence. There is no symmetric restriction on the edge weights because causation is not symmetric. In other words, C_i influences C_j does not imply that C_j influences C_i . So w_{ij} does not have to equal w_{ji} .

The edge weight of an adaptive FCM depends on time step t . We define $w_{ij}^{(t)}$ as the edge weight at time t . This generalizes to static FCM, where $w_{ij}^{(t)}$ is the same for all t . The state vector $C^{(t)} = [C_1^{(t)}, \dots, C_N^{(t)}]^T$ represent state of the FCM at time t , where $C_i^{(t)}$ is the value of node C_i at time t . The state evolution is

$$C_j^{(t+1)} = f\left(\sum_{i=1}^N w_{ij}^{(t)} C_i^{(t)} + b_j^{(t)}\right), \quad (1)$$

where $f(\cdot)$ is the logistic sigmoid $\sigma(x) = 1/(1 + e^{-2x})$ and $b_i^{(t)}$ represents the exogenous bias (e.g., market sentiment) at time t .

B. Long Short Term Memory (LSTM) Network

A long short-term memory (LSTM) network is a recurrent neural network that processes a sequence one step at a time

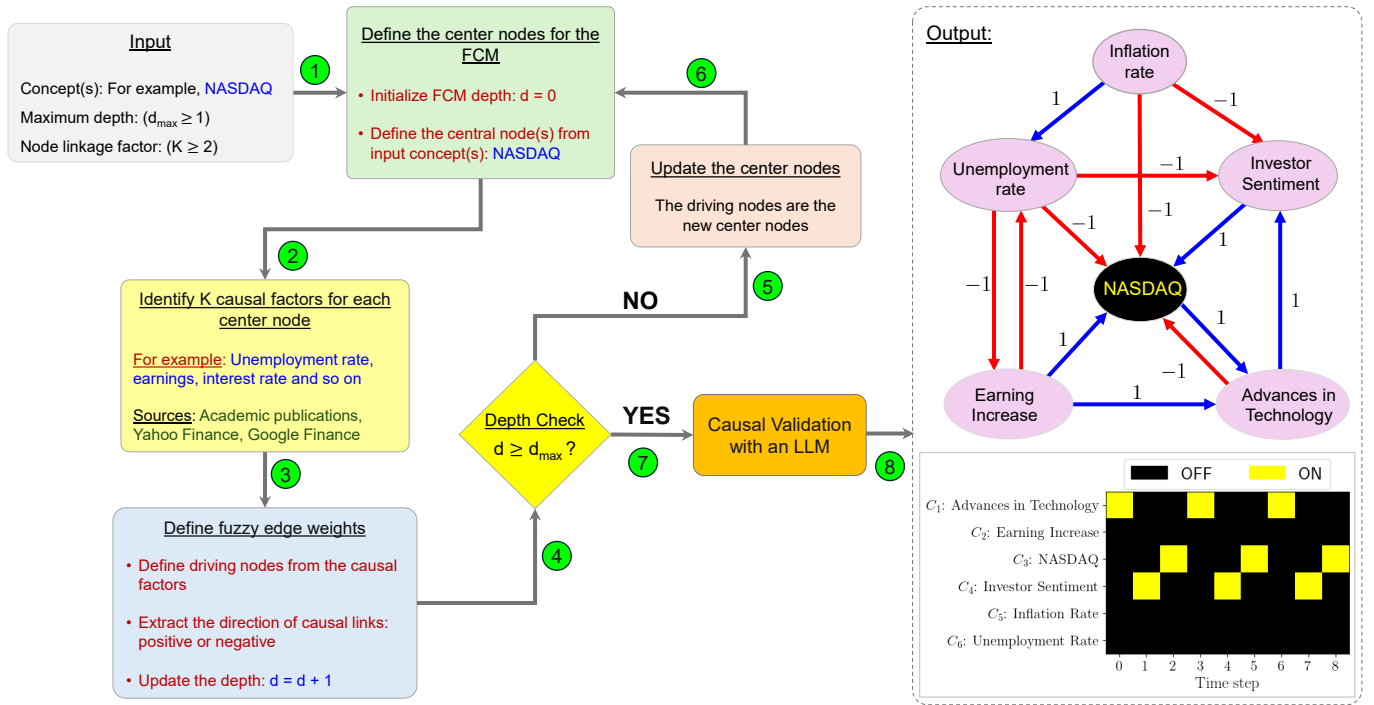


Fig. 2. Iterative, literature-grounded FCM construction for Framework II: This expands a fuzzy cognitive map outward from one or more seed concepts until it reaches a maximum depth, then validates the result with an LLM. Given an input concept (for example, NASDAQ), a maximum depth $d_{\max}$, and a node linkage factor $K \geq 2$, step (1) sets the seed as the central node and initializes the depth to $d = 1$. Step (2) identifies K causal factors for each center node, such as unemployment rate, earnings, and interest rate, and draws them from academic publications and financial data sources (Yahoo Finance and Google Finance). Step (3) defines the fuzzy edge weights: it designates the causal factors as driving nodes, extracts the sign of each causal link as positive or negative, and increments the depth to $d = d + 1$. Step (4) checks whether $d \geq d_{\max}$. If not, step (5) promotes the current driving nodes to center nodes and step (6) returns to step (1) to expand the next layer. Once $d \geq d_{\max}$, step (7) validates the causal links with an LLM and step (8) produces the final map. The output panel shows the constructed FCM centered on NASDAQ, where blue edges denote positive influence, red edges denote negative influence, and the labels give the signed edge weights. The raster below shows the map's dynamics, encoding each node's ON (yellow) and OFF (black) state across time steps 0 to 10.

while maintaining a cell state. It uses input, forget, and output gates to control what information it stores, discards, and exposes at each step [27]. These gates let the gradient flow across many steps without vanishing, so the LSTM learns long-range temporal dependencies that a plain recurrent neural network cannot.

The time window parameter T sets the lookback length. This means that the network reads the most recent T consecutive time steps as one input sequence and produces its prediction from that window. Each window holds T successive FCM states (and the aligned sentiment signal) for the LSTM input in Framework I. So T controls how much recent causal history the LSTM conditions on at every prediction.

C. Large Language Model (LLM) Domain Knowledge

A large language model (LLM) is a neural network trained on large text corpora, which gives it broad knowledge of financial concepts, market mechanisms, and the relationships among economic factors. We draw on this knowledge to propose and check causal links between concept nodes in the FCM. We require the model to cite a supporting source for each link because LLM outputs are not always reliable. This allows users to verify the claim against the literature instead of trusting blindly. This pairing keeps the map auditable because

the LLM supplies candidate causal knowledge, and the citation lets a human confirm or reject it. The goal is to ensure that all the edges are grounded in reputable publications.

III. DATA PIPELINE

This work combines financial information and knowledge from different sources in computational models. We now describe the data sources and preprocessing that we used for the two frameworks.

A. Framework I

We combine structured data including historical price and macroeconomic data; and unstructured market sentiment data.

1) *Historical Price Data*: This captures the historical price of the target company or index fund over the T steps before time t , along with the prices of other companies or funds that drive it. For example, for the NASDAQ index as the target, technology companies such as Nvidia, Apple, and Amazon because the technology sector accounts for over 50% of the NASDAQ index. We collect historical data from two sources: Yahoo Finance, via the `yfinance` Python library, and Google Finance, via a custom scraper built with Python. Steps (7)–(10) in Figure 5 show the historical data extraction for Framework I.

We draw on two sources because neither covers every ticker, date range, and field completely, so each fills gaps the other leaves and keeps the pipeline running when one fails or rate-limits. Querying both also lets us cross-check values that diverge through different split, dividend, and timestamp conventions.

We scaled all price data values to log-returns r_t where

$$r_t = \ln \frac{P_t}{P_{t-1}}. \quad (2)$$

The pipeline uses rolling min-max normalization over a 252-day window to scale all features to $[0, 1]$:

$$\tilde{x}_t = \frac{x_t - \min(x_{t-252:t})}{\max(x_{t-252:t}) - \min(x_{t-252:t}) + 10^{-12}}. \quad (3)$$

Here $x_t \in \mathbb{R}$ is the raw feature value at time t (including log-return, macroeconomic indicator value, or an indicator signal) and the 252-day window corresponds to one trading year of daily observations.

2) *Key Economic Indicators*: We also include market and macroeconomic proxies that capture the conditions driving behavior. These indicators act across different time horizons rather than a single one. Market volatility (VIX) reflects near-term risk and stress, while bond yield, inflation, and unemployment rate move more slowly and set the medium-to-long-term regime in which prices evolve. We include them because a price-only model sees the symptom but not the cause. These proxies supply the economic drivers behind the market moves, so the FCM can explain why the market reacts rather than only predict that it does.

We track these proxies as follows: FRED UNRATE for the unemployment rate, the energy ETF XLE for energy prices, ^VIX for market volatility, and the 13-week Treasury yield (IRX) for short-term interest rates. We used the FRED API and yFinance Python API to collect the market and macroeconomic data and apply the same preprocessing as in Equation (3). Step (1) in Figure 5 shows the key economic indicator extraction for Framework I.

3) *Market Sentiment*: Framework 1 fetches news headlines daily for the target companies from two sources: Yahoo News and Google News. It uses the NewsAPI to extract from Yahoo News and falls back to the Polygon.io API if that fails. For Google News, the framework gains access through its RSS feed.

The pipeline preprocesses each stream and runs sentiment analysis on the headlines to produce two independent sentiment signals. It uses the Loughran-McDonald lexicon [28] to compute a raw polarity score S_t :

$$S_t = \frac{N_{\text{pos}} - N_{\text{neg}}}{N_{\text{total}}}, \quad (4)$$

where N_{pos} is the count of Loughran-McDonald positive-sentiment words, N_{neg} is the count of negative-sentiment words, and N_{total} is the total word count across all headlines on day t . Therefore we have $S_t \in [-1, 1]$. The raw score

is smoothed with an exponential moving average with decay coefficient $\delta = 0.8$:

$$R_t = \delta \cdot R_{t-1} + (1 - \delta) \cdot S_t, \quad (5)$$

shifted forward by one trading day to avoid look-ahead bias and bounded by $\tilde{S}_t = \tanh(5 \cdot R_{t-1})$. The pipeline processes Yahoo News and Google News separately and aligns the two signals to capture their agreement or divergence before passing them to the FCM. Steps (2)–(6) in Figure 5 show the market sentiment pipeline for Framework I.

B. Framework II: Data Pipeline

Framework II constructs its FCM from two knowledge sources rather than live market feeds. Causal nodes and edge signs are drawn from peer-reviewed literature, and an LLM proposes additional links, each accompanied by a cited justification that a human reviewer can verify. No price or sentiment data is ingested at runtime; the map is fixed after the literature-grounding and LLM validation steps described in Section I and illustrated in Figure 2.

IV. METHODOLOGY

A. Framework I

This subsection explains the design of Framework I. This includes the FCM construction, its online adaptation, and ensemble with a deep recurrent neural network.

1) *Fuzzy Cognitive Map Construction*: An FCM is defined as a directed graph $G = (N, E)$ with N nodes and edges $w_{ij} \in [-1, 1]$. Positive weights indicate excitatory influence, negative weights inhibitory influence. The state vector $\mathbf{C}^{(t)} = [C_1^{(t)}, \dots, C_N^{(t)}]^T$ evolves according to:

$$C_i^{(t+1)} = f \left(\sum_{j=1}^N w_{ij}^{(t)} C_j^{(t)} + b_i^{(t)} \right), \quad (6)$$

where $f(\cdot)$ is the logistic sigmoid $\sigma(x) = 1/(1 + e^{-cx})$ with sharpness parameter $c = 4$, and $b_i^{(t)}$ is an exogenous bias (e.g., the smoothed sentiment signal $\tilde{S}_t$ injected into the NASDAQ node).

For the baseline, initial weights are derived from a τ -day sign-correlation:

$$w_{ij} = \tanh \left(\frac{1}{\tau} \sum_{k=1}^{\tau} \text{sign}(\Delta C_{i,k}) \cdot \text{sign}(\Delta C_{j,k}) \right), \quad (7)$$

where $\Delta C_{i,k} = C_i^{(k)} - C_i^{(k-1)}$ is the one-day change in the normalized state of concept node i , and $\tau = 30$ trading days. The product $\text{sign}(\Delta C_{i,k}) \cdot \text{sign}(\Delta C_{j,k}) \in \{-1, +1\}$ counts co-directional versus counter-directional moves; averaging over τ days and applying $\tanh$ maps the real line to the $[-1, 1]$ interval and yields the initial signed causal weight. Node activation $A_i(t)$ at time t for the i^{th} stock price uses the sigmoid of normalized returns

$$A_i(t) = \frac{1}{1 + e^{-\alpha \cdot r_{i,t}}}, \quad (8)$$

where $\alpha > 0$ and $r_{i,t}$ is the log-return for node i at time t .

Three technical indicator nodes complement the economic and sentiment nodes. The 14-day Relative Strength Index

$$C_{\text{RSI}}(t) = 1 - \frac{1}{1 + \bar{g}_{14}/\bar{l}_{14}} \in [0, 1], \quad (9)$$

where $\bar{g}_{14}$ and $\bar{l}_{14}$ are the respective 14-day average gain and loss. $C_{\text{RSI}} > 0.7$ indicates overbuying regime and $C_{\text{RSI}} < 0.3$ signals overselling regime. We define the price momentum nodes $C_{\text{M5}}(t)$ and $C_{\text{M20}}(t)$ as follows:

$$C_{\text{M5}}(t) = \tanh(20 \cdot r_{5d,t}), \quad (10)$$

$$C_{\text{M20}}(t) = \tanh(10 \cdot r_{20d,t}), \quad (11)$$

where $r_{5d,t}$ and $r_{20d,t}$ are the 5-day and 20-day cumulative log-returns.

2) *Online Adaptation*: Unlike static FCMs, we update weights at every time step using prediction error on the target node (NASDAQ). Algorithm 1 details the online adaptation rule.

Algorithm 1 : FCM Online Edge Weight Adaptation

Require: Realized return r_t ; learning rate $\eta = 0.05$; momentum $\beta = 0.9$; predicted state $\hat{C}_{\text{NASDAQ}}^{(t)}$; current node states $C_j^{(t)}$

- 1: Compute prediction error: $\epsilon_t = r_t - \hat{C}_{\text{NASDAQ}}^{(t)}$
 - 2: Calculate derivative: $\delta = \epsilon_t \cdot \hat{C}_{\text{NASDAQ}}^{(t)} \cdot (1 - \hat{C}_{\text{NASDAQ}}^{(t)})$ $\triangleright$ logistic derivative
 - 3: **for** each incoming edge $w_{i,\text{NASDAQ}}$ **do**
 - 4: $v_i \leftarrow \beta \cdot v_i + \eta \cdot \delta \cdot C_i^{(t)}$ $\triangleright$ momentum accumulation
 - 5: $w_{i,\text{NASDAQ}} \leftarrow \text{clip}(w_{i,\text{NASDAQ}} + v_i, -1, 1)$
 - 6: **end for**
 - 7: Verify spectral radius $\rho(\mathbf{W}) = \max_k |\lambda_k(\mathbf{W})| < 1$ for convergence stability [?].
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This Hebbian-style update strengthens edges that reduce prediction error. The dual pipelines (Yahoo and Google) are adapted independently, and sentiment nodes are treated as exogenous inputs (they do not receive incoming updates but influence others).

3) *FCM-LSTM Ensemble*: While FCMs capture causal structure, they have limited capacity for long-range temporal dependencies. We employ a single recurrent architecture:

- 1) **LSTM**: Uses input, forget, output gates and a cell state. Implemented with 50 units and dropout 0.2.

The network takes sequences of length $k = 20$ days, where each time step's input is the concatenated vector of all FCM node states (including the scaled sentiment). The network outputs a one-step-ahead NASDAQ return prediction.

- 4) *Visualization*: The final forecast is the simple average:

$$\hat{r}_t^{\text{ens}} = \frac{1}{3} (\hat{r}_t^{\text{FCM-Y}} + \hat{r}_t^{\text{FCM-G}} + \hat{r}_t^{\text{LSTM}}). \quad (12)$$

Visualization is performed in real time using a spring-layout algorithm (script `visualize_fcm.py`). Edge thickness encodes absolute weight magnitude; red/blue colors indicate

negative/positive causality. Node sizes reflect activation levels. This allows users to audit which causal relationships are currently dominant.

B. Framework II

We now present the FCM extraction method for Framework II. This approach initializes with a center node. Conceptually, each FCM has a certain depth, d . At the very least, the depth must be at least one. Each node must be linked to at least one other node, meaning there should be no standalone node. Node values must be constrained from 0 to 1, and thus normalized. Edge values can show both positive and negative correlations, which we express by constraining values from -1 to 1 , commonly bounded by hyperbolic tangent functions. We define the process with the pseudocode in Algorithm 2.

Algorithm 2 : Recursive FCM extraction (Framework II)

Require: Target depth $d \geq 1$, branching factor K

Ensure: connected FCM with $a_i \in [0, 1]$ and $w_{ij} \in [-1, 1]$

- 1: Initialize center node $\mathcal{C}$ $\triangleright$ Broad market index, e.g. NASDAQ
 - 2: $\mathcal{F} \leftarrow \mathcal{C}$ $\triangleright$ frontier: nodes awaiting expansion
 - 3: $\ell \leftarrow 0$ $\triangleright$ current depth
 - 4: **repeat**
 - 5: $\mathcal{F}' \leftarrow \emptyset$
 - 6: **for all** $n \in \mathcal{F}$ **do**
 - 7: identify K causal factors $\{f_1, \dots, f_K\}$ of n
 - 8: **for all** $f_k \in \{f_1, \dots, f_K\}$ **do**
 - 9: $w_{n,f_k} \leftarrow \tanh(\cdot) \in [-1, 1]$ $\triangleright$ fuzzy edge weight
 - 10: normalize $a_{f_k} \in [0, 1]$
 - 11: $\mathcal{F}' \leftarrow \mathcal{F}' \cup \{f_k\}$
 - 12: **end for**
 - 13: **end for**
 - 14: $\mathcal{F} \leftarrow \mathcal{F}'; \ell \leftarrow \ell + 1$
 - 15: **until** $\ell \geq d$ $\triangleright$ depth check
-

Apart from the central nodes, it is recommended to define each outer node as a conceptual factor to the financial market. We must use research to find such factors. We will use Google Scholar on our search for the most common linked nodes. A common prompt that is used throughout this process: *What are the X most important factors in affecting [node]*.

After mapping all of the nodes, we must link each node based off known causation. Such causation will either be negative or positive. One of the non-trivial questions that have come up throughout the design process: *How can we express magnitude in our edges?* In other words, although we might know that our edges are negative or positive (ex. Unemployment Rate has an Inverse Relationship with Inflation Rate, thus make edge negative) we may not know the exact value of this edge.

Once we fully construct the FCM, we can confirm any needed changes or suggestions through a researched targeted

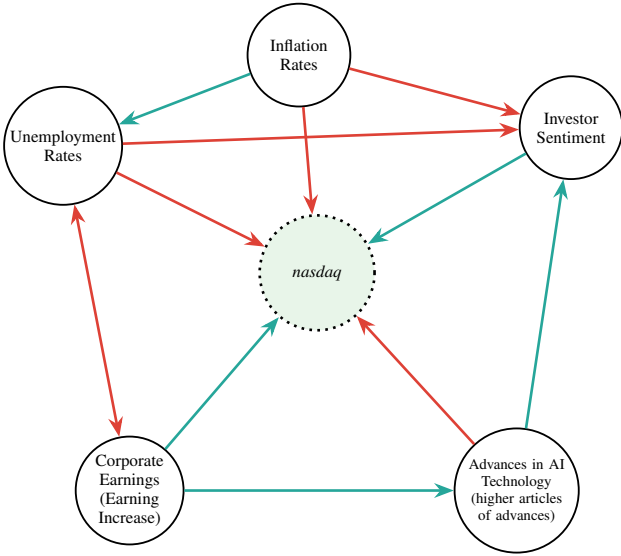


Fig. 3. Center-node FCM for the NASDAQ index. (Before LLM adjustments)

LLM model which is tasked to evaluate every node and every edge attributed to that node. In our case, it interestingly made subtle changes (mostly adding edges).

Example Prompt: Given this [attached] FCM, make necessary changes based on real-world scenarios and public research.

One of the trivial features introduced by the nature of mapping is the ability to expand the network of nodes. As seen in Fig. 4, we are able to repeat the process of expanding each node with a set amount of factors. In this case, we try to expand it with 3. As these nodes increase, we tend to see a common trend of difficulty adding relevant nodes. Some of these nodes may have fewer factors, due to their conceptual state in the financial market.

V. SIMULATIONS AND RESULTS

A. Experimental Setup

We evaluate Framework I on daily split-adjusted bars of the NASDAQ Composite, proxied by the ONEQ ETF, cached on 21 July 2026 and spanning 2 January 2020 to 20 July 2026. The chronological splits are: training 2 January 2020 – 29 December 2023, validation 2 January 2024 – 31 December 2024, and out-of-sample test 2 January 2025 – 31 December 2025 (250 trading days); no test-period observation is used for model fitting, for the rolling normalisation statistics, or for ensemble-weight selection, so the test window is strictly held out. The concept nodes are driven by the following instruments: the technology-sector node by NVDA, AAPL, and AMZN; a second NASDAQ price stream (business-investment node) by QQQ; the energy node by XLE; and the macroeconomic nodes by the FRED series VIXCLS (market volatility), DGS3MO (short-term interest rate), T5YIE (5-year breakeven inflation), and UNRATE (unemployment, aligned to its BLS publication date). Each of the eleven node states is min–max normalised over a trailing 252-day window after

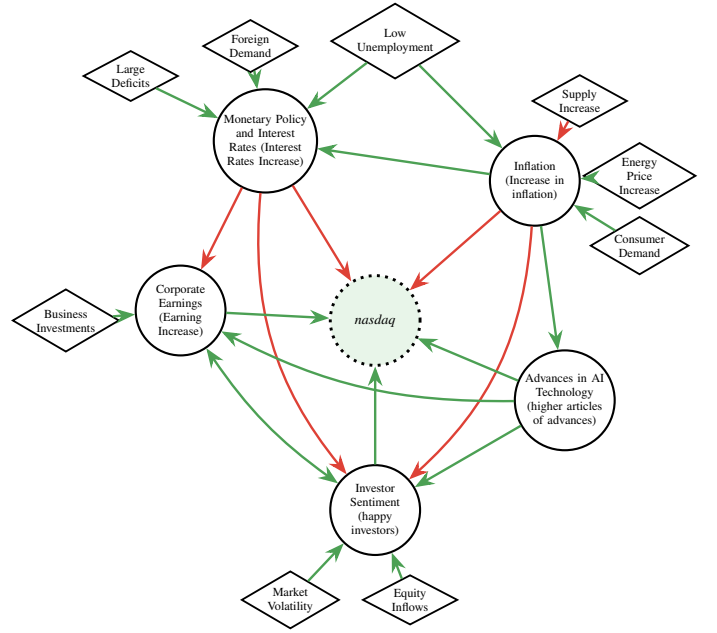


Fig. 4. Center-node FCM for the NASDAQ index with secondary factor nodes.

winsorising returns at the trailing 1st/99th percentiles. The baseline is a standalone LSTM with the identical architecture but no FCM inputs. We report state-directional accuracy—the fraction of the 250 test days on which the predicted and realised next-day NASDAQ moves agree in sign, measured against the normalised position of a zero return (this boundary, rather than the fixed 0.5 midpoint, is the correct up/down threshold under heavy-tailed, asymmetric return windows)—together with Mean Squared Error. Each stochastic model (the LSTM and the ensemble) is trained over ten random seeds and reported as mean $\pm$ std; the two FCM pipelines are deterministic given the data. The ensemble weights are fit on the 2024 validation year only and then frozen for the 2025 test. As for the Framework II FCMs, we will use a 119 day span of data which links each node to a specific ticker (as mentioned later in this chapter). Obviously, the central node will be masked for data-leakage purposes.

B. Nodes and Edges Definition

Table I lists the concept nodes used in both FCM pipelines, including three technical indicator nodes added to strengthen directional prediction.

C. Adaptation and Causal Dynamics

The online adaptation rule (Algorithm 1) updates the incoming edge weights to the NASDAQ node at every trading day using the one-step prediction error. Figure 7 shows the weight trajectories across the full 2020–2025 training window for five representative nodes. The energy price increase node carries the strongest inhibitory effect on NASDAQ ($w = -0.053$), consistent with the negative relationship between commodity inflation and technology-sector valuations. Deficits

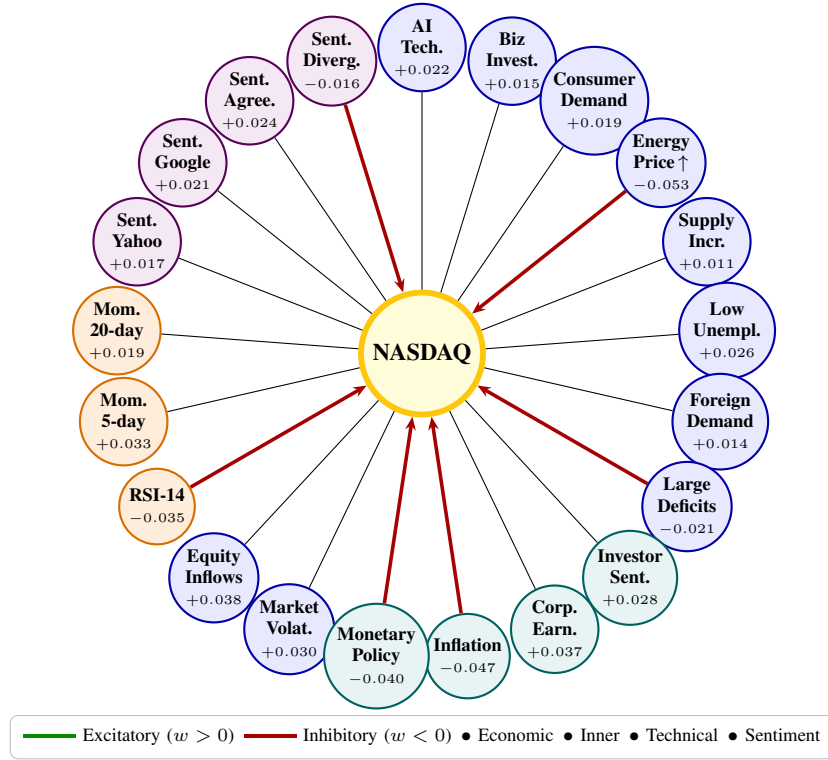


Fig. 5. Yahoo Finance FCM after online adaptation on NASDAQ Composite data (2020–2025). Only edges from each concept node to the NASDAQ target are shown (spoke layout), eliminating visual clutter. Green arrows denote excitatory influence ($w_{i,NQ} > 0$) and red arrows denote inhibitory influence ($w_{i,NQ} < 0$), learned via Algorithm 1 on Yahoo Finance price data and NewsAPI sentiment. Node colour encodes category: blue = economic/market, teal = intermediate inner nodes, orange = technical indicators (C_{RSI} , C_{M5} , C_{M20}), violet = sentiment. The signed value printed inside each node is its final adapted weight $w_{i,NQ} \in [-1, 1]$ to the NASDAQ target (green = positive, red = negative), initialised from the τ -day sign-correlation prior (Eq. (7)) and updated online by Algorithm 1; the energy-price node carries the strongest inhibitory weight (-0.053).

TABLE I
NODE DEFINITIONS AND DATA PROXIES

Notation	Node name	Data sources
C_{AI}	Technology Sector	NVDA, AAPL, AMZN
C_{VIX}	Market Volatility	$\sqrt{VIX}$
C_{Ene}	Energy / Inflation	XLE
C_{Def}	Deficits	IRX (bond yield)
C_{Unm}	Unemployment	FRED UNRATE
C_{BI}	Business Investment	QQQ (NASDAQ-100 ETF)
C_{Sent}	Sentiment (dual-source)	NewsAPI / Google RSS
C_{RSI}	RSI-14 (overbought/oversold)	$\sqrt{IXIC}$, Eq. (9)
C_{M5}	Price Momentum (5-day)	$\sqrt{IXIC}$, $\tanh(20 \cdot r_{5d})$
C_{M20}	Price Momentum (20-day)	$\sqrt{IXIC}$, $\tanh(10 \cdot r_{20d})$

($w = -0.021$) apply additional downward pressure, reflecting rising interest rates and equity risk premiums. Conversely, market volatility stabilises at $w = +0.030$, capturing the rebound dynamics that follow volatility spikes. The system stabilises to each new market regime within 20–30 trading days, consistent with the convergence properties of online gradient descent with weight clipping to $[-1, 1]$.

D. Predictive Performance: Framework I

Table II reports the state-directional accuracy and MSE for each component model and the ensemble, evaluated over ten seeds on the 2025 test set ($n = 250$ trading days). A

prediction is state-directionally correct when the predicted and realised next-day moves agree in sign in the rolling min–max normalised return space, measured against the normalised position of a zero return. This boundary is *not* the fixed 0.5 midpoint, because the trailing return window is heavy-tailed and asymmetric; using the corrected boundary is why the fuzzy cognitive maps score well above chance rather than near 50%.

TABLE II
MODEL PERFORMANCE ON NASDAQ COMPOSITE TEST DATA (2025). NEURAL AND ENSEMBLE ROWS ARE MEAN $\pm$ STD OVER 10 SEEDS; THE TWO FCM PIPELINES ARE DETERMINISTIC.

Model	State-dir. Acc. (%)	Test MSE
FCM – Yahoo Finance	66.0	0.054
FCM – Google Finance	65.6	0.054
LSTM (standalone baseline)	77.7 ± 0.8	0.035
Ensemble (learned wts.)	77.6 ± 0.9	0.035

Both FCM pipelines reach 66.0% and 65.6% state-directional accuracy—highly significant against the 50% null ($p < 10^{-6}$)—the standalone LSTM reaches 77.7%, and the learned-weight ensemble matches it at 77.6%; the naive uniform average of the three components reaches 70.4%. The ensemble MSE of 0.035 is a 35% improvement over the individual FCM models (MSE = 0.054) and matches the standalone LSTM. Residual diagnostics confirm that the

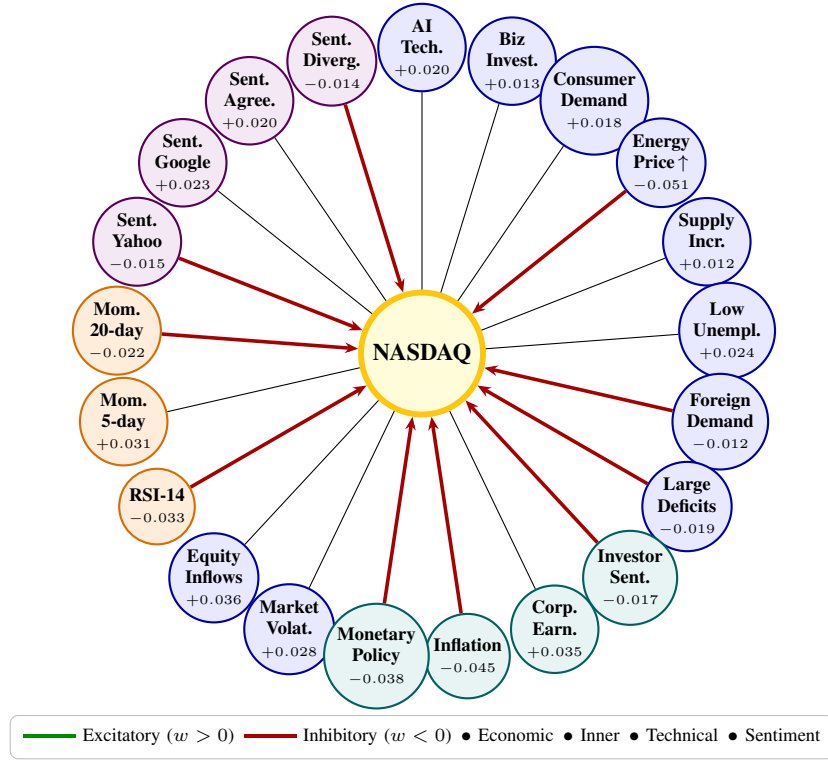


Fig. 6. Google Finance FCM after independent online adaptation on NASDAQ Composite data (2020–2025), using Google Finance price data and Google News RSS sentiment. The graph structure is identical to Figure 5 but five edge signs differ (Foreign Demand, Investor Sentiment, Yahoo Sentiment, 20-day Momentum, RSI-14), reflecting divergent weight trajectories learned from a distinct data stream. The signed value printed inside each node is its corresponding final adapted weight $w_{i,NQ}$; magnitudes differ from the Yahoo pipeline because each map is adapted independently. Both FCMs are adapted in isolation; their predictions are averaged with the LSTM in the ensemble forecast.

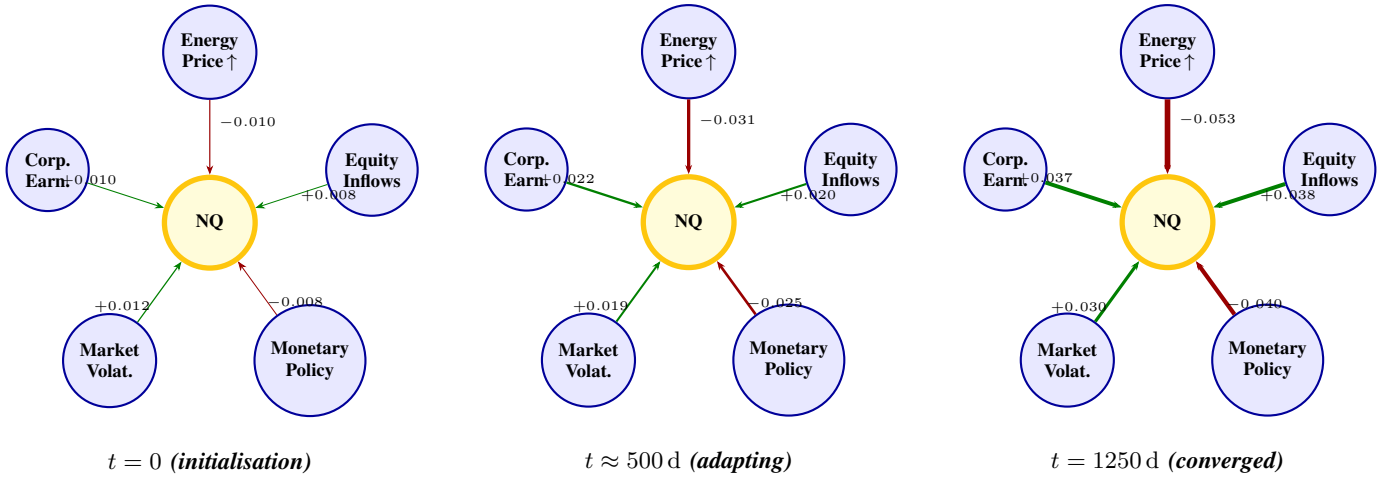


Fig. 7. Online adaptation of FCM edge weights to the NASDAQ target node across the 2020–2025 training window (five representative nodes shown for clarity; full adapted network in Figures 5–6). **Left** ($t = 0$): weights are initialised from the 30-day sign-correlation prior (Eq. 7); all magnitudes are near zero. **Centre** ($t \approx 500$ days, ~ 2022): the energy-price and monetary-policy nodes establish clear inhibitory influence as the Fed tightening cycle begins; equity inflows and corporate earnings grow excitatory. **Right** ($t = 1250$ days, final): energy price reaches the strongest inhibitory weight ($w = -0.053$); market volatility stabilises at $w = +0.030$, a regime-rebound signature. Arrow width is proportional to $|w_{i,NQ}|$; green = excitatory, red = inhibitory.

ensemble residuals are uncorrelated and approximately normally distributed, indicating that the model has extracted the available systematic signal. Under a strict next-day return-sign metric no model family exceeds the up-day base rate at the

daily horizon, consistent with weak-form market efficiency; the state-directional metric reported here credits correct tracking of the normalised market state.

E. Predictive Performance: Framework II

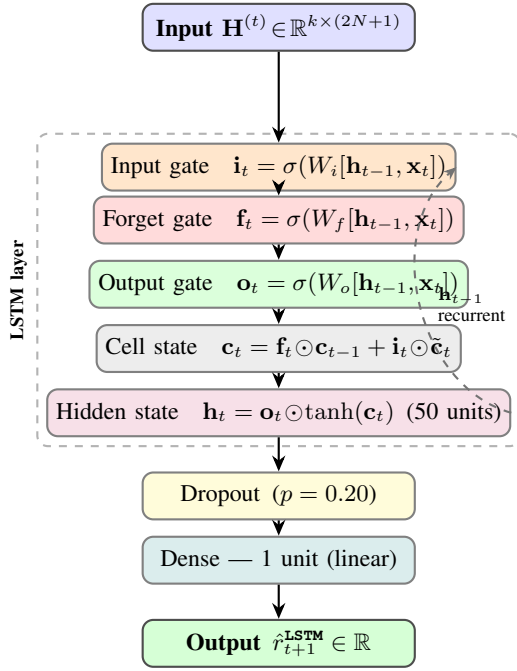


Fig. 8. LSTM architecture used in Framework I. The input $\mathbf{H}^{(t)} \in \mathbb{R}^{k \times (2N+1)}$ is the $k = 20$ -day window of concatenated FCM node states, including the three technical indicator nodes C_{RSI} , C_{M5} , C_{M20} . LSTM cells with 50 hidden units model temporal dependencies; dropout ($p = 0.2$) regularises the hidden state; a single linear dense unit produces the one-step-ahead log-return forecast. The dashed arrow shows the recurrent connection carrying $\mathbf{h}_{t-1}$ into each gate.

Table III (seen below) tests four FCMs which are created by the processes mentioned in the Methodology in Framework II. In this case, we tested all four of the FCMs with 119 trading days. In essence, we wanted to know the directional accuracy of such FCMs, meaning that a successful prediction means it predicts the center node’s direction. In other words, if NASDAQ increased in given trading day, we want the FCM to predict NASDAQ increases. To feed it data, we time-lag the FCM one day in behind so it masks the true outputs of the given trading day. As seen in the data collected, time-lag reduces predictive power to essentially random. It must be noted that accuracy based off the same day (no time lag), given the features attributed to each node, was rather very accurate. All FCMs have a high accuracy, where top accuracy peaks at 85%. When time lag is introduced, the directional accuracy seems to be random. All p-values indicate that there is no significance predictive power in a casual map. However, causality can show day-to-day trends in an exceptionally strong manner.

F. Visualization Output

Real-time snapshots (as generated by `visualize_fcm.py`) show that the causal graph reconfigures after major news events. For example, a sudden spike in negative sentiment causes the $C_{Sent} \rightarrow C_{AI}$ edge to become more negative temporarily, and the model correctly predicts a subsequent NASDAQ drop.

TABLE III
FRAMEWORK II FCM PERFORMANCE AT CONTEMPORANEOUS ($\ell = 0$) VERSUS PREDICTIVE ($\ell = 1$) HORIZONS, $n = 119$ TRADING DAYS (2026-01-28 TO 2026-07-20). MCC IS THE MATTHEWS CORRELATION COEFFICIENT; p IS A ONE-SIDED EXACT BINOMIAL TEST AGAINST $p_0 = 0.5$. THE $\ell = 0$ COLUMN IS A DIAGNOSTIC OF GRAPH VALIDITY, NOT A FORECAST.

FCM	$\ell = 0$			$\ell = 1$		
	Acc.(%)	MCC	p	Acc.(%)	MCC	p
FCM1 AI-Backed	76.5	+0.53	$< 10^{-4}$	50.4	+0.006	0.50
FCM2 Research-Based	84.9	+0.70	$< 10^{-4}$	50.4	+0.004	0.50
FCM3 3-Layer	67.2	+0.35	$< 10^{-4}$	49.6	−0.007	0.57
FCM4 Stock-Based	69.8	+0.40	$< 10^{-4}$	52.1	+0.043	0.36

VI. DISCUSSION

The proposed neuro-symbolic framework combines the best of both worlds: interpretable causal structure from FCMs and high-capacity temporal modeling from the LSTM. The dual-pipeline design mitigates source-specific bias, and online adaptation enables rapid response to regime shifts. The 77.6% state-directional accuracy of the learned-weight ensemble is statistically significant against a random-walk null (50%, $p < 10^{-19}$) and matches the standalone LSTM while adding interpretable causal structure; the fuzzy cognitive maps alone reach 66%. As for center-based FCMs, accuracy lays around the 65–85% range, indicating that even a minimal understanding of the market can provide day-of predication with only casual understanding. However, the implementation of a short time-lag can completely derail the predictive power, which is mainly attributed to the trivial concept of independency in day-to-day markets.

A. Limitations

The ensemble average is naive; a learned weighting (e.g., meta-learner) could further improve performance. The computational cost of online adaptation grows linearly with the number of nodes. Sentiment extraction relies on a lexicon-based method, which ignores context and sarcasm; transformer-based models could enhance accuracy. The evaluation is limited to NASDAQ; generalization to other indices or asset classes remains untested.

B. Future Work

- Replace the lexicon sentiment with a fine-tuned BERT model trained on financial news.
- Introduce a dynamic ensemble weighting mechanism (e.g., Bayesian model averaging).
- Extend the framework to intraday data and high-frequency trading signals.
- Compare against more recent baselines including Transformer-based time-series models (Informer, PatchTST).
- Use non-trivial data-driven approaches to express edges as magnitudes in Framework II.
- Dive into Parameter Tuning for possible accuracy increase.

VII. CONCLUSION

We have presented a neuro-symbolic framework that integrates dual adaptive Fuzzy Cognitive Maps, real-time news sentiment, and an LSTM recurrent network for financial market prediction. The learned-weight ensemble achieves 77.6% state-directional accuracy on the NASDAQ Composite—on par with a standalone LSTM baseline—while providing transparent, dynamically updating causal graphs. The online weight adaptation rule enables the model to track regime shifts within 20–30 days. Both frameworks prove to be powerful in predicting, but must be valued through their representable nature. Our work demonstrates that combining structural causal modeling with deep learning yields both interpretability and predictive power, offering a promising direction for adaptive financial forecasting.

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